

Volume Survey Report

Jharia Block-7 · Dhanbad, Jharkhand

Report Date	22 May 2026, 06:29 UTC	Mine Type	Opencast Coal
Drone Model	DJI Matrice 300 RTK	Resolution	3.2 cm/px
Images Processed	47	GCPs Used	8
GCP RMSE	±2.5 cm	Coordinate System	WGS84 / UTM Zone 44N

Volume Analysis Results

Metric	Value	Unit
Stockpile Volume	1,894,397	m ³
Cut Volume	3,372,027	m ³
Fill Volume	1,477,630	m ³
Net Change	+54,421	m ³
Mine Area	2.4	km ²
Maximum Pit Depth	47.3	m
Elevation Range	183.0 – 231.0	m ASL

Processing Pipeline

Step	Status	Detail
Image Alignment	✓ Complete	47/47 images · GCP-corrected
DEM / DSM Generation	✓ Complete	1m resolution · 8.2M point cloud
Orthomosaic	✓ Complete	3.2 cm/px RGB GeoTIFF
Volume Calculation	✓ Complete	1m contour interval · TIN method
Report Generation	✓ Complete	PDF + Excel exports

This report was generated by MineVisionAI. Volume calculations use TIN-based DEM differencing. Results are accurate to ±1% under normal flight conditions with adequate GCP distribution.